GENERATIVE AI COURSE

Module 3: Generative AI Tools &
Real-World Applications

Study Notes • Practical Examples • Knowledge Check

Learning focus
Understand AI tool categories, generate digital content, explore multimodal AI, complete real-world tasks, and verify AI outputs responsibly.

Prepared as a beginner-friendly learning resource

1. Introduction to Generative AI

Generative Artificial Intelligence (Generative AI) is a type of AI that can create new content from instructions called **prompts**. Depending on the system, it can produce text, images, presentations, computer code, audio, video, and other digital content.

Basic workflow

INPUT — User Prompt
Question, instruction, image, or file

AI PROCESSING
The model interprets the input and generates a response

OUTPUT — Generated Content
Text, image, code, presentation, audio, or video

2. Categories of Generative AI Tools

Text generation

Examples: ChatGPT, Claude, Google Gemini.

Applications: articles, explanations, summaries, emails, reports, and study notes.

Output: written content.

Image generation

Examples: Adobe Firefly, Midjourney, ChatGPT image generation.

Applications: posters, illustrations, concept art, product concepts, and visual assets.

Output: AI-generated images.

Presentation generation

Examples: Canva and Microsoft PowerPoint features.

Applications: slide decks, educational presentations, business pitches, and project reports.

Output: presentation slides.

Code generation

Examples: GitHub Copilot, ChatGPT, Cursor.

Applications: writing code, explaining programming, finding bugs, and creating tests.

Output: code and technical explanations.

Audio and voice generation

Examples: ElevenLabs and Suno.

Applications: narration, speech, sound effects, and music.

Output: audio content.

Video generation

Examples: Runway and Adobe Firefly features.

Applications: short clips, animations, advertisements, and visual stories.

Output: video content.

3. What Is Multimodal AI?

Multimodal AI can process or generate more than one kind of information, such as text, images, audio, and video. The exact capabilities depend on the tool.

Example: Upload a photograph of a science experiment, ask AI to explain it, request a diagram, and prepare presentation slides.

Common modalities

Text	Questions, answers, summaries
Images	Visual analysis and creation
Audio	Speech, sound, transcription
Video	Scenes, motion, video summaries

4. Real-World Applications

Field	Application	Example output
Education	Explain topics and support practice	Study notes and quizzes
Software development	Assist with programming	Code and tests
Business	Prepare and organize information	Business proposal
Marketing	Create promotional content	Social media advertisement
Healthcare	Support information and documentation	Draft education material for professional review
Cybersecurity	Explain defensive security	Security awareness checklist
Engineering	Explore design concepts	Prototype concept
Entertainment	Create stories and media	Story, music, or video
Government	Simplify public information	Citizen information guide
Research	Summarize and organize sources	Research outline

> Review AI-generated results carefully when accuracy, safety, or important decisions are involved.

5. Practical Exercise: Future Technology Company Presentation

Goal: Produce a short presentation introducing an AI technology company.

Step 1 — Generate text
Prompt: “Write a five-slide presentation introducing a future AI technology company, its mission, products,

applications, and roadmap.”

Step 2 — Generate an image
Prompt: “Create an original futuristic AI research campus with robotics, holographic displays, and clean-energy technology.”

Step 3 — Create slides
Use a presentation tool to arrange the text, images, titles, and diagrams into five slides.

Step 4 — Generate code (optional)
Ask an AI coding assistant to create a simple HTML landing page for the company.

Step 5 — Verify the output
Check facts, spelling, image rights, code functionality, and whether the presentation meets the project requirements.

6. Prompt Engineering: Input and Output

A good prompt clearly explains the task, context, requirements, and desired output format.

> **Example input prompt**
> “Explain Generative AI to a beginner in simple English. Include its definition, five applications, advantages, limitations, and one practical example. Present the answer using headings and a summary table.”

Expected output

- Clear definition of Generative AI
- Five practical applications
- Advantages and limitations
- Real-world example
- Summary table

7. Responsible AI and Fact Verification

AI can generate incorrect facts, outdated information, misleading images, or code containing bugs. Verify important outputs before relying on them.

Accuracy: Check important claims against reliable sources.

Privacy: Avoid uploading passwords, confidential files, or sensitive personal information unnecessarily.

Bias: Consider whether the output is unfair or incomplete.

Copyright: Respect rights and licenses of content used or generated.

Security: Test AI-generated code and do not run untrusted code blindly.

Transparency: Disclose AI assistance when required.

Human review: Make final decisions yourself or involve an appropriate expert.

8. Knowledge Check

1. What is the main purpose of Generative AI?

A. Only store information
B. Create new content from learned patterns and prompts
C. Only calculate numbers

Answer: B

2. Which is an example of multimodal AI?

A. A tool that handles text and images
B. A basic calculator only
C. A folder of text files

Answer: A

3. Which tool category assists with writing computer programs?

A. Code generation
B. Image compression only
C. Spreadsheet formatting only

Answer: A

4. What should you do before using important AI-generated facts?

A. Trust everything automatically
B. Verify them using reliable sources
C. Delete all references

Answer: B

5. What is a prompt?

A. An instruction or input given to an AI system
B. A computer monitor
C. A type of battery

Answer: A

9. Final Learning Outcomes

After completing Module 3, a learner should be able to:

1. Identify different categories of Generative AI tools.
2. Generate text, images, presentations, and code using suitable prompts.
3. Explain multimodal AI and its uses.
4. Select AI tools for education, business, software development, and content creation.
5. Complete a practical task using multiple AI tools.
6. Verify AI-generated information and apply responsible AI practices.

> **Final module output:** A practical mini-project containing AI-generated text, one visual, a short presentation, and a verification checklist.